



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SAKTHI AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

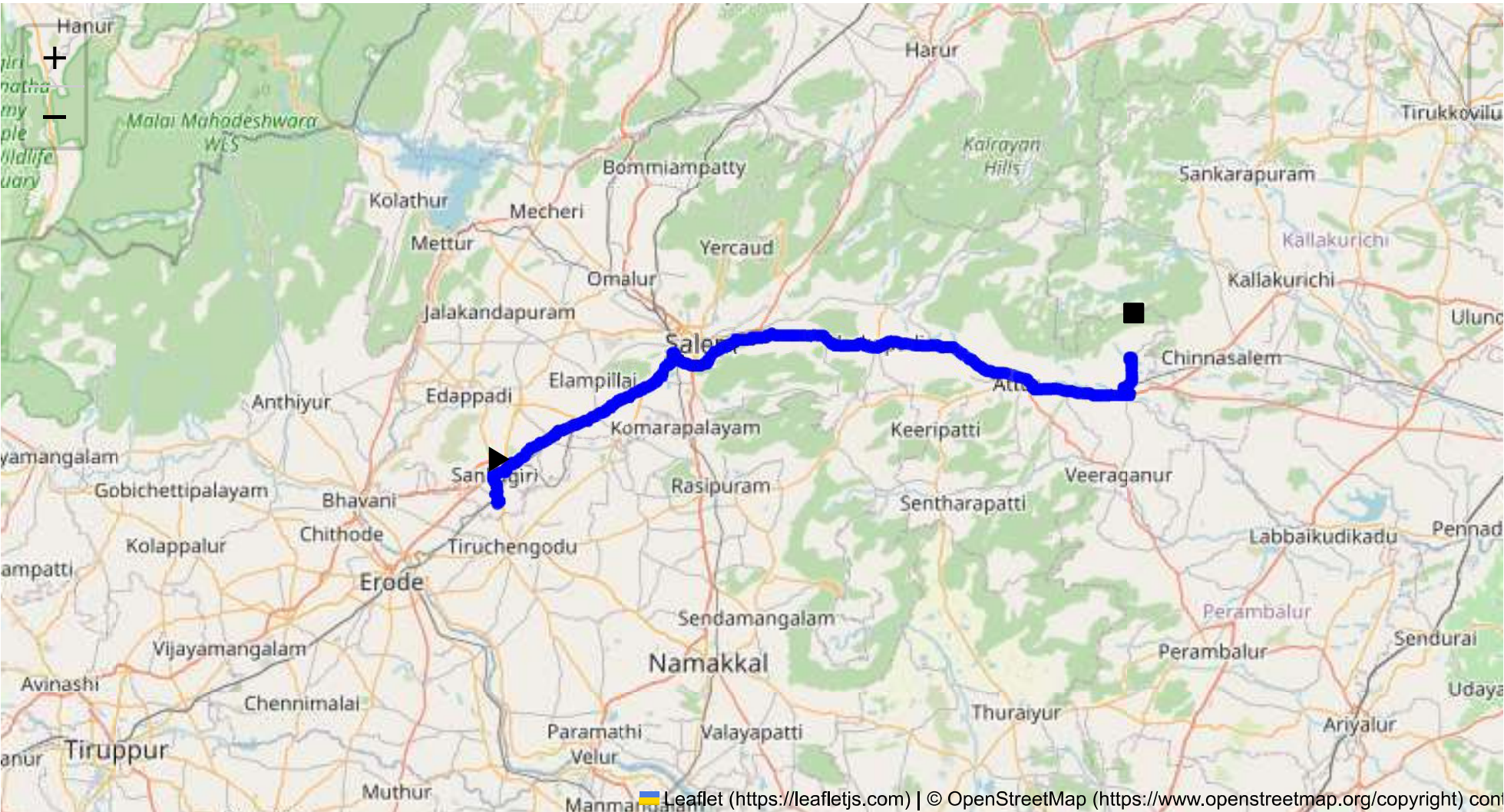
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 120.53 km
Estimated Duration: 2.3 hours
Adjusted Duration (Heavy Vehicle): 2.9 hours
Start: (11.4381, 77.8734)
End: (11.636297, 78.754288)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

- The route from Sangagiri, Tamil Nadu to Siruvachur, Tamil Nadu is approximately 120.53 kilometers long.

- Key highways and roads involved likely include segments of NH 44 and NH 79, given typical driving paths in the region.
- The route traverses mostly rural and semi-urban areas, with some stretches through towns and small cities.

2. Weather Conditions and Hazards

- **Typical Weather:** The region generally experiences a tropical climate with hot summers and moderate to heavy monsoon rains from June to September.
- **Hazards:** The monsoon season can lead to waterlogged roads and reduced visibility, increasing the risk of hydroplaning and accidents. Heat can also cause road surface issues with bitumen.

3. Traffic Patterns

- **Peak Hours:** Expect heavier traffic during morning (8 AM to 10 AM) and evening (5 PM to 8 PM) hours, especially near towns.
- **Congestion-Prone Areas:** Small urban centers along the route, such as town junctions, often experience congestion during peak times and market days.

4. Road Quality and Infrastructure

- **Assessment:** Main highway sections are typically well-maintained but expect varying quality on secondary roads.
- **Potential Issues:** Potholes, uneven surfaces, and narrow segments in rural areas can pose challenges. Roadworks are also common.

5. Alternative Routes for Emergencies

- Alternative routes include deviations through parallel state highways or rural roads, though these may be longer and of lesser quality.
- It's advisable to consult real-time map data for any closures or significant delays.

6. Local Regulations on Hazardous Material Transport

- Tamil Nadu has regulations mandating specific licenses and clear labeling for vehicles transporting hazardous materials.
- Restrictions may apply on certain vehicles within city limits during peak hours.

7. Historical Incidents

- Some incidents involving heavy vehicles in the region include rollovers and collisions, often due to inadequate safety precautions or driver fatigue.
- Specific data on hazardous material incidents may be limited but historically tied to driver error or mechanical failure.

8. Environmental Considerations

- The route passes through agricultural land; hence, spills can severely impact local water bodies and soil.
- Avoid sensitive ecological areas or bypass them using prescribed routes.

9. Communication Coverage

- **Coverage:** Generally strong near urban centers but potential dead zones in rural stretches.
- **Provider Variability:** Different service providers may offer varying levels of coverage.

10. Emergency Response Times

- Response times can vary: urban areas may see quicker responses (20-30 minutes), while rural segments could take longer (30-60 minutes).
- Closest major hospitals and fire stations are generally located in larger towns along the route.

12. Overall Summary of Risk Assessment

- **Risks:** Weather issues during monsoon, road quality variability, congestion near towns, potential for communication dead zones.
- **Mitigation:** Plan travel during non-peak hours, regularly check vehicle conditions especially during monsoon, and maintain continuous communication with support teams.
- **Preparation:** Always have emergency contact numbers available and inform local authorities about schedule and cargo for smooth operation along regulated zones.

By thorough planning and regular communication, the risk associated with this route for heavy vehicles can be mitigated effectively. Regular updates and route monitoring are essential for ensuring safety and compliance.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	Medium	11.45000, 77.87213	30 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr
6	Turn	Medium	11.45309, 77.87325	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
7	Turn	Medium	11.46400, 77.87323	30 KM/Hr
8	Turn	Medium	11.46765, 77.87037	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.49149, 77.89601	15 KM/Hr
11	Turn	High	11.49224, 77.89601	15 KM/Hr
12	Turn	Medium	11.53501, 77.95707	30 KM/Hr
13	Turn	Medium	11.53601, 77.95856	30 KM/Hr
14	Turn	Medium	11.64172, 78.11817	30 KM/Hr
15	Turn	Medium	11.64216, 78.11875	30 KM/Hr
16	Turn	High	11.64195, 78.12129	15 KM/Hr
17	Turn	High	11.64120, 78.12168	15 KM/Hr
18	Turn	Medium	11.63948, 78.12081	30 KM/Hr
19	Turn	Medium	11.65703, 78.20465	30 KM/Hr
20	Turn	High	11.58641, 78.75308	15 KM/Hr
21	Turn	High	11.58802, 78.75186	15 KM/Hr
22	Turn	High	11.58887, 78.75197	15 KM/Hr
23	Turn	High	11.59101, 78.74641	15 KM/Hr
24	Turn	Medium	11.59140, 78.74636	30 KM/Hr
25	Turn	Medium	11.59195, 78.74592	30 KM/Hr
26	Turn	High	11.59227, 78.74527	15 KM/Hr
27	Turn	Medium	11.59301, 78.74506	30 KM/Hr
28	Turn	High	11.59483, 78.74584	15 KM/Hr
29	Turn	High	11.59462, 78.74629	15 KM/Hr
30	Turn	Medium	11.59756, 78.75122	30 KM/Hr
31	Turn	High	11.60684, 78.75495	15 KM/Hr
32	Turn	Medium	11.60701, 78.75487	30 KM/Hr
33	Turn	High	11.60730, 78.75420	15 KM/Hr
34	Turn	Medium	11.63286, 78.75400	30 KM/Hr
35	Turn	High	11.63321, 78.75339	15 KM/Hr
36	Turn	Medium	11.63354, 78.75336	30 KM/Hr
37	Turn	High	11.63626, 78.75403	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
21	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
22	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
23	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
24	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
25	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
26	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
27	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
28	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
29	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
30	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
34	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
36	clinic	Lavanya Clinic	11.652121, 78.4795146	30 km/h	Medium
37	hospital	AmsH Agathiyar Sidha Hospital	11.650564, 78.483542	30 km/h	Medium
38	clinic	Sudha Polyclinic	11.647825, 78.508537	30 km/h	Medium
39	hospital	Government Hospital, Kothampadi	11.62641, 78.542199	30 km/h	Medium
44	clinic	Sugam Clinic	11.5832481, 78.7496438	30 km/h	Medium
45	hospital	Selvam Hospital	11.585978, 78.748761	30 km/h	Medium
46	hospital	Prabushankar Hospital	11.584239, 78.7490323	30 km/h	Medium
47	hospital	Pannerselvam Hospital	11.587014, 78.755058	30 km/h	Medium
48	clinic	Sri Shakthi Clinic	11.5868967, 78.7555902	30 km/h	Medium
49	hospital	Dr. Prabhusankar Hospital	11.586292, 78.755837	30 km/h	Medium
50	hospital	Meenachi Hospital	11.6370665, 78.7556702	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
14	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
20	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
31	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
32	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
33	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
35	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium
40	school	Vani Vikas Matric Hr.sec school	11.6212713, 78.5462658	30 km/h	Medium
41	school	Saraswati Metric Hr sec School	11.5910827, 78.6402264	30 km/h	Medium
42	college	Maruthi Polytechnic College	11.5856852, 78.7201072	30 km/h	Medium
43	college	Bharathiyar Educational Institure	11.5863463, 78.7364793	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

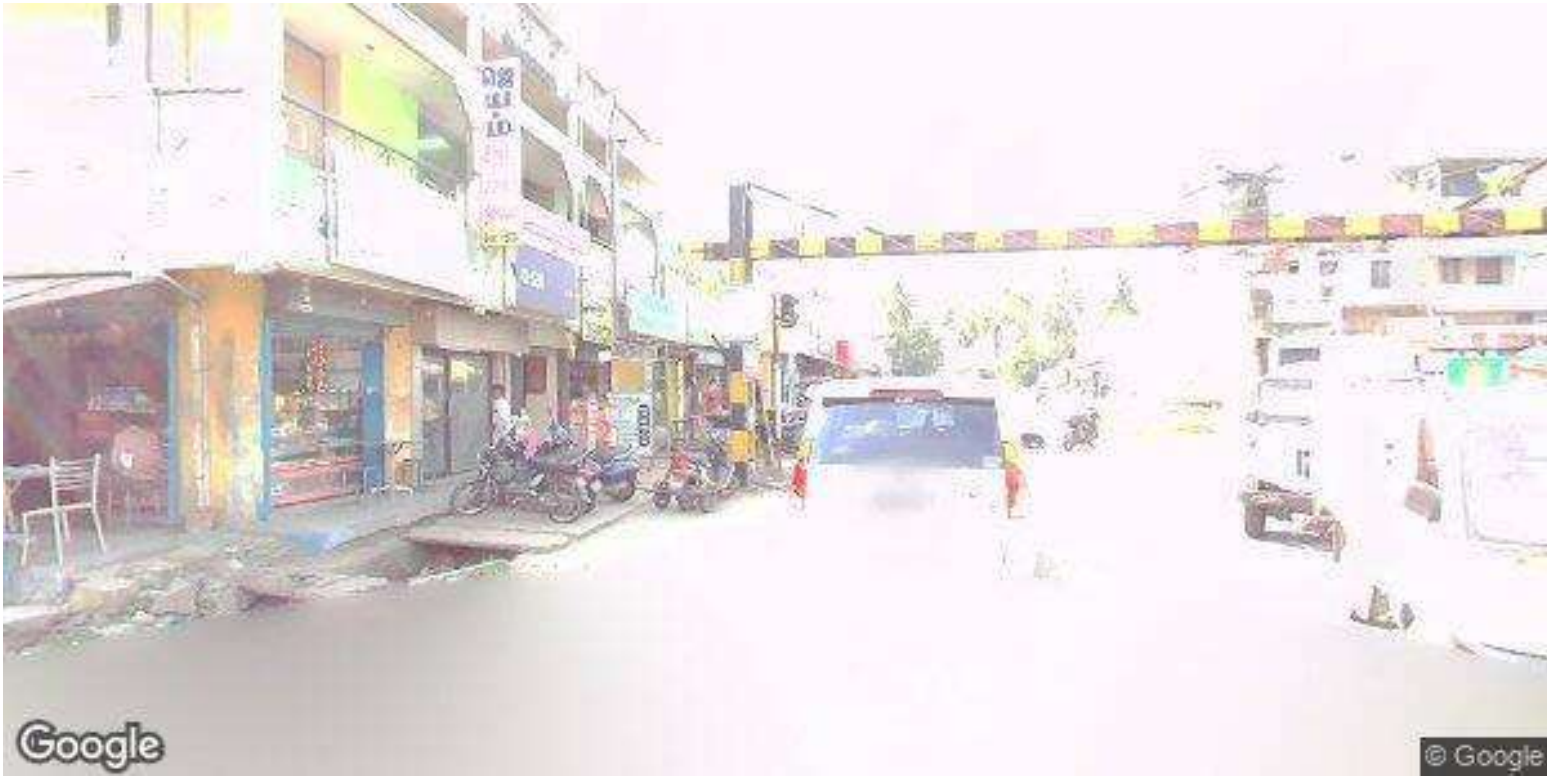


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45000, 77.87213



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45309, 77.87325



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64172, 78.11817



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64216, 78.11875



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64195, 78.12129



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64120, 78.12168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63948, 78.12081



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65703, 78.20465



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58641, 78.75308



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58802, 78.75186



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58887, 78.75197



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59101, 78.74641



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59140, 78.74636



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59195, 78.74592



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59227, 78.74527



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59301, 78.74506



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59483, 78.74584



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59462, 78.74629



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59756, 78.75122



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60684, 78.75495



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60701, 78.75487



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60730, 78.75420



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63286, 78.75400



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63321, 78.75339



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63354, 78.75336



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63626, 78.75403

